Next level of reproducible, comparable and integrable Metabolomics: An ELIXIR Implementation Study

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ELIXIR Metabolomics Community

Motivation

The ELIXIR metabolomics community relies on standards, formats and data treatment solutions development and adoption, but challenges remain in 1) ensuring high-quality reported metadata, 2) sufficiently contextualised results, 3) interoperable and reusable datasets and 4) integrating these metabolomics data with other omics or other studies.

How can we improve existing data & metadata models to represent **semantics** to better integrate and enrich existing and new data?

How can we improve **data quality** and support **quality control** in data acquisition, data processing and storage?

How can we **compare** metabolome composition and quantities **across multiple studies, organisms and tissues** and interpret their differences?

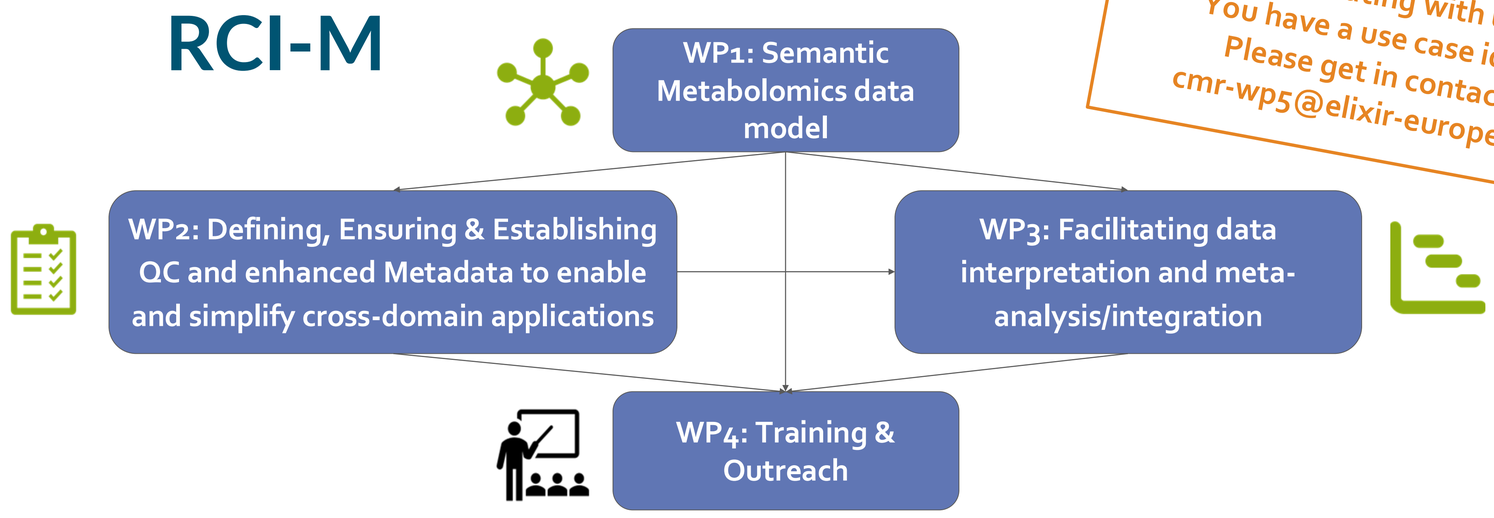
How can we integrate with **other omics** datasets to answer integrated research questions in a systems biology context?

Participation is still possible!
Your community is interested in collaborating with us?
You have a use case idea?
Please get in contact!
cmr-wp5@elixir-europe.org

Semantic Metabolomics

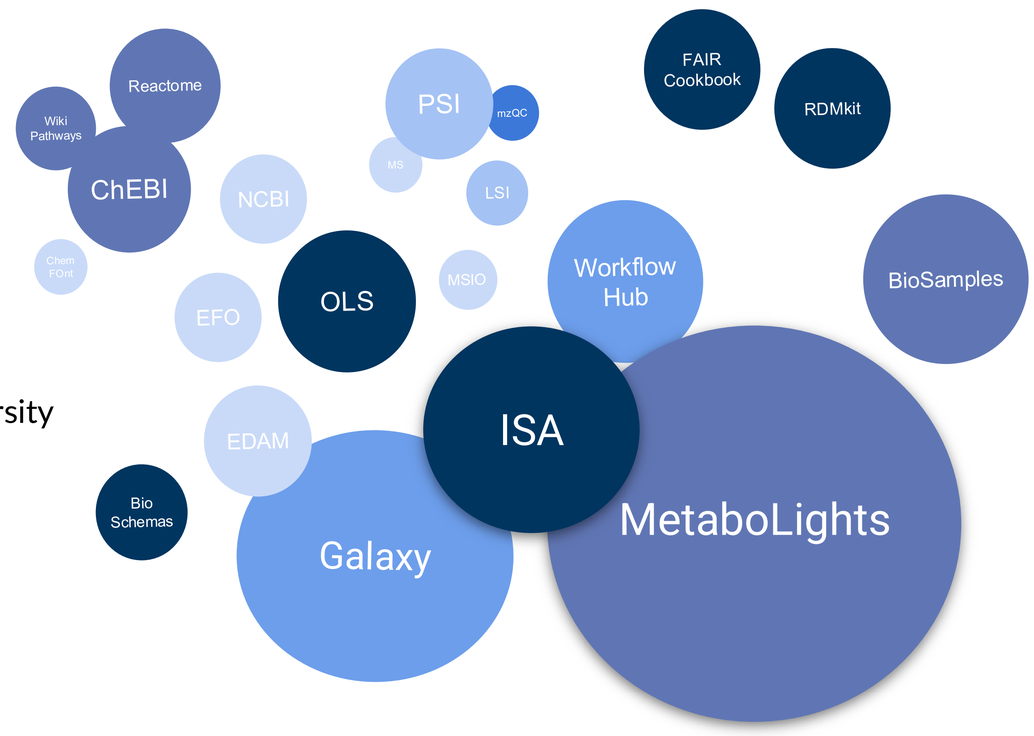
- Applying Ontologies and controlled vocabularies to describe metabolomics data
- Support machine readability, integration, and FAIR principles
- Enables harmonization across studies, species, and domains
- Integrating Ontologies, first-order logic and Semantic Web
- Essential for ML/AI applications and multi-modal data integration

RCI-M



Currently 19 participants from 9 ELIXIR nodes (CZ, DE, EMBL, ES, FR, IT, NL, SE, UK)
Many more are part of the regular community meetings

Supporting communities: *Metabolomics, Proteomics, Galaxy, Toxicology, RDM*
Collaborating with platforms: *Interop, Tools, Data*



Opportunities

- ELIXIR offers a unique and beneficial framework for trans-disciplinary research
- Key-standards / formats and ELIXIR services
- FAIRification from the point of view of metabolomics
- Added-value of the semantic technologies (Ontologies, knowledge graphs, ...)

1 EMBL-EBI
2 INRAE - MetaboHUB
3 INRAE
4 Translational Genomics, Maastricht University
5 Cajal Institute
6 RECETOX, Masaryk University
7 IBG-5, Forschungszentrum Jülich GmbH
8 University of Groningen
9 IBIOM - CNR
10 Uppsala University
11 Leibniz Institute of Plant Biochemistry

ELIXIR is an intergovernmental organization that brings together life science resources from across Europe. The scientific and technical activities in ELIXIR are run by five Platforms and a number of Communities.

The goal of the ELIXIR Metabolomics Community is to work with experimental scientists and developers to provide the resources, analysis tools and infrastructure that will help metabolite identification.

https://elixir-europe.org/communities/metabolomics

These projects have received funding from the European Union's Horizon 2020 research and innovation programme

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